

Technology Stack

Date	04 July 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive, user-friendly interface for farmers, researchers, and policymakers. Bootstrap ensures mobile compatibility and faster UI development.

2	Backend / Server-Side	Python (Flask)	Flask is lightweight, easy to integrate with Machine Learning models, and provides fast processing of crop recommendation requests.
3	Database / Data Storage	MySQL	Stores user information, crop datasets, prediction history, and environmental data with reliable relational database management.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere <i>(or AWS if available)</i>	Enables easy deployment, scalability, and online accessibility of the application with minimal maintenance.
5	Version Control & CI/CD	Git & GitHub	Facilitates version control, team collaboration, source code management, and project backup.

6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib	Scikit-learn is used for crop prediction using machine learning. Pandas and NumPy handle data preprocessing and analysis. Matplotlib is used for visualization, and Joblib stores the trained ML model for prediction.